Student Performance & Academic Analytics

Exploratory Data Analysis Report

Mathematics and Portuguese Student Datasets

Tools: Python • Pandas • NumPy • Matplotlib • Seaborn

1. Introduction

This project performs exploratory data analysis (EDA) on the UCI Student Performance datasets for Mathematics and Portuguese. The analysis studies students' final grades (G3) and explores how academic history, study habits, family background, and selected lifestyle variables are associated with performance.

2. Objectives

- Inspect the datasets and assess data quality.
- Explore numerical and categorical variables using descriptive statistics and visualizations.
- Examine relationships between student characteristics and final grades.
- Compare patterns between Mathematics and Portuguese.
- Summarize findings and suggest practical educational actions.

3. Dataset Description

The data source is the UCI Machine Learning Repository's Student Performance dataset. The Mathematics dataset contains 395 rows and 33 columns; the Portuguese dataset contains 649 rows and 33 columns. The outcome used throughout this analysis is G3, the final grade on a 0–20 scale. The data includes demographic, family, school, social, and academic attributes.

Dataset	Rows	Columns	Target
Mathematics	395	33	G3
Portuguese	649	33	G3

4. Data Quality and Preparation

- Both datasets were inspected for dimensions, data types, missing values, duplicate records, and expected value ranges.
- The checks reported no missing values and no exact duplicate rows in either dataset.
- No rows were removed during the validation stage; the analysis proceeded with the original observations.

5. Exploratory Analysis

The notebook includes distributions of final grades and age, boxplots for grade distributions and group comparisons, bar charts for categorical variables, count plots, scatter plots, and correlation heatmaps. The analyses cover study time, previous failures, previous period grades, parental education, family size, parents' status, gender, extracurricular activities, absences, and other available variables.

6. Key Findings

- **Previous grades:** G1 and G2 show the strongest positive correlations with G3 in both subjects. G2–G3 correlation is approximately 0.905 for Mathematics and 0.919 for Portuguese.
- **Past failures:** The number of previous class failures is negatively associated with final grade (correlation approximately –0.360 in Mathematics and –0.393 in Portuguese). Group averages also show lower final grades among students with recorded failures.
- **Study time:** Study time is positively associated with G3 in Portuguese (correlation approximately 0.250) and weakly associated in Mathematics (approximately 0.098). Group averages do not increase perfectly across every study-time category.

- **Absences:** The relationship with G3 is weak in both datasets (approximately 0.034 in Mathematics and −0.091 in Portuguese). Absence count alone is not a strong indicator of final grade in this analysis.
- **Grade distributions:** Mathematics G3 has mean 10.42 and median 11; Portuguese G3 has mean 11.91 and median 12. Mathematics grades are more dispersed in this sample.
- **Gender:** Group averages differ between female and male students in the two subject datasets, but these descriptive differences should not be interpreted as causal or generalized beyond the sample.

7. Recommendations

- Use previous grades as one early indicator to identify students who may benefit from academic support.
- Offer targeted follow-up and learning support to students with previous class failures.
- Encourage consistent study routines and evaluate support using subsequent progress.
- Avoid using a single attribute—such as absences, gender, or family background—to make judgments about an individual student.
- Interpret results alongside student context and other educational evidence.

8. Limitations

- This is observational EDA; correlations and group differences do not establish causation.
- The datasets represent particular schools and may not be representative of all student populations.
- Some categories have small sample sizes, making their averages less stable.
- The available columns do not capture every factor that can influence academic performance.

9. Conclusion

The EDA indicates that students' earlier academic grades and previous class failures are more strongly associated with final grades than several demographic or lifestyle variables examined. Study time shows a clearer positive association in Portuguese than in Mathematics, while absences have weak relationships with G3. These findings can inform further investigation and supportive interventions, but should be used carefully and not as causal conclusions.

10. Reproducibility

The analysis is provided in the Jupyter notebook *student_performance_analysis.ipynb*. Place the Mathematics and Portuguese CSV files alongside the notebook, ensure their filenames match the paths used in the code, and select **Kernel** → **Restart & Run All** to reproduce the analysis.

Dataset source: UCI Machine Learning Repository — Student Performance
https://archive.ics.uci.edu/dataset/320/student+performance